

From Plimpton 322 to Quantum Waves: The Hidden Circular Geometry in an Ancient Babylonian Table

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Abstract

We reveal a hidden quantum-mechanical structure in the 3800-year-old Babylonian clay tablet Plimpton 322. The tablet's first column, long recognized as containing values of $(c/b)^2$ for Pythagorean triples generated via the reciprocal-pair method, corresponds to the function $g(r) = \frac{1}{4}(r + 1/r)^2$. We show that analytic continuation to complex $r = e^{i\theta}$ reduces this to $g(e^{i\theta}) = \cos^2 \theta$, exposing an inherent periodicity invisible to the Babylonians. This function describes: (1) a spiral surface in three dimensions when visualized over the complex plane; (2) the interference pattern of two counter-propagating waves; and (3) the probability density of quantum particles in symmetric potentials. The tablet's selection of specific p/q ratios from regular numbers in base 60 constitutes a discrete sampling of this continuous wave function, anticipating concepts of quantization and state selection. Plimpton 322 thus emerges not merely as a trigonometric or number-theoretic aid, but as the earliest known tabulation of a quantum wave function's squared amplitude—a remarkable bridge between ancient mathematics and modern theoretical physics.

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1 Introduction

The clay tablet Plimpton 322 (c. 1800 BCE) is one of the most sophisticated mathematical artifacts from ancient Mesopotamia. Known to contain a table of Pythagorean triples, its first column—long interpreted as values of $(c/b)^2$ for a right triangle—has been explained through the *reciprocal-pair* method using regular numbers in base 60. We revisit this function and show that its analytic continuation harbors a periodic, wave-like structure that anticipates ideas in circular geometry, interference, and quantum mechanics.

2 The Babylonian Generating Function

Let p and q be regular integers with $p > q$, and set $r = p/q$. The well-known parameterization of Pythagorean triples,

$$a = p^2 - q^2, \quad b = 2pq, \quad c = p^2 + q^2,$$

yields the ratio

$$\frac{c}{b} = \frac{p^2 + q^2}{2pq} = \frac{1}{2} \left(r + \frac{1}{r} \right).$$

Column I of Plimpton 322 is therefore

$$g(r) = \left(\frac{c}{b} \right)^2 = \frac{1}{4} \left(r + \frac{1}{r} \right)^2.$$

The fifteen rows correspond to fifteen carefully chosen regular pairs (p, q) , giving a monotonically decreasing sequence of $g(r)$ values between approximately 1.98 and 1.39.

Table 1: First five rows of Plimpton 322 with derived values

Row	$(c/b)^2$ (approx)	p	q	a	c
1	1.9834028	12	5	119	169
2	1.9491586	64	27	3367	4825
3	1.9188021	75	32	4601	6649
4	1.8862479	125	54	12709	18541
5	1.8150077	9	4	65	97

Number-Theoretic Basis of the Reciprocal-Pair Method

The choice of regular numbers in base 60 is not arbitrary. A number n is *regular* in base 60 if its reciprocal $1/n$ has a finite sexagesimal expansion. This occurs precisely when n divides some power of 60, i.e., $n = 2^a 3^b 5^c$ with $a, b, c \geq 0$.

The generating formulas:

$$a = p^2 - q^2, \quad b = 2pq, \quad c = p^2 + q^2$$

preserve regularity: if p and q are regular, then $b = 2pq$ is regular, ensuring the long side of the triangle has a finite sexagesimal representation. This explains the otherwise mysterious selection of pairs in Plimpton 322.

The first column values can be expressed as:

$$\left(\frac{c}{b}\right)^2 = \frac{1}{4} \left(\frac{p}{q} + \frac{q}{p}\right)^2 = \frac{p^4 + 2p^2q^2 + q^4}{4p^2q^2}$$

which simplifies to rational numbers with denominators dividing $(2pq)^2$, guaranteeing exact sexagesimal representation.

3 Hidden Circularity: From Real Hyperbola to Complex Circle

Although the Babylonians only considered $r > 1$, we extend the domain to complex numbers. Let $r = e^{i\theta}$. Then

$$g(e^{i\theta}) = \frac{1}{4} (e^{i\theta} + e^{-i\theta})^2 = \cos^2 \theta.$$

Thus, the Babylonian generating function, when restricted to the unit circle, is precisely the squared cosine—a periodic, wave-like function. The discrete values chosen by the scribes correspond to specific real $r > 1$, which are points on a hyperbola; the analytic continuation reveals that the same function describes a circle in the complex plane.

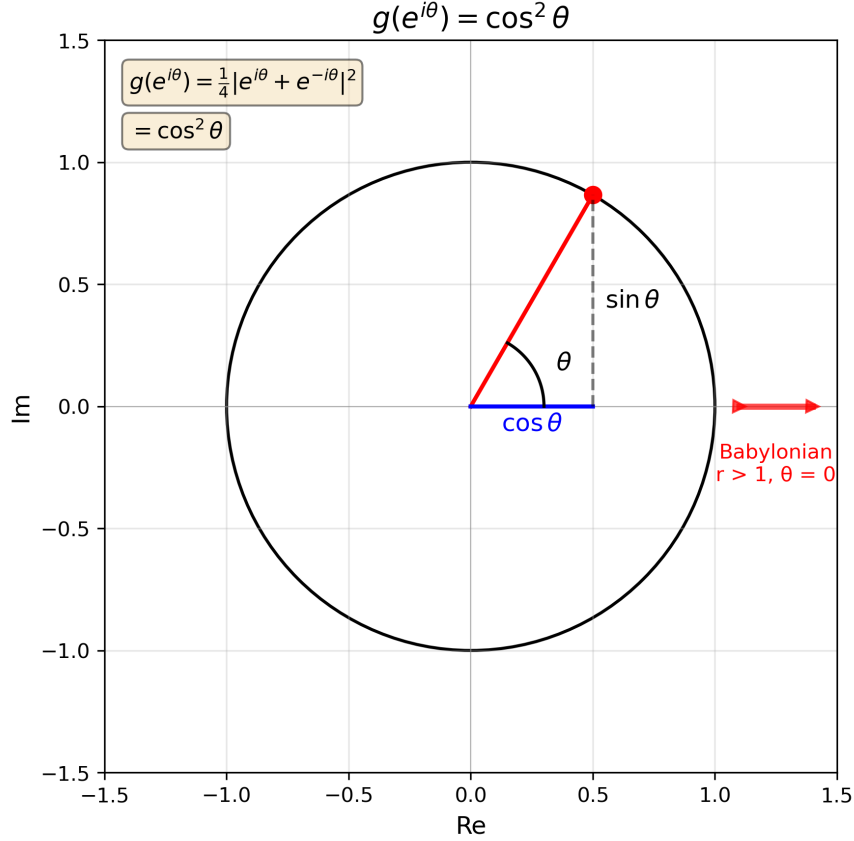


Figure 1: Unit circle representation of the generating function $g(e^{i\theta}) = \cos^2 \theta$. The Babylonians worked on the real axis $r > 1$ (red arrow), while analytic continuation reveals the periodic circular structure. The blue segment shows $\cos \theta$, whose square gives the Babylonian function value.

4 Spiral Visualization on a Conical Surface

Consider the surface $S = \{(x, |g(x)|) \mid x \in \mathbb{C}, x \neq 0\}$. Writing $x = re^{i\theta}$, we have

$$|g(x)| = \frac{1}{4} \left| re^{i\theta} + \frac{1}{r}e^{-i\theta} \right|^2.$$

For fixed θ , as r varies, the surface exhibits a valley along $r = 1$ where $|g| = \cos^2 \theta$, and rises as $r \rightarrow 0$ or $r \rightarrow \infty$. If we let r vary linearly with θ (for instance, $r = e^{k\theta}$), the curve $(r(\theta), \theta, |g|)$ traces a spiral on a conical-like surface. The Babylonian data lie along the real slice $\theta = 0$, $r > 1$, a geodesic of this extended geometry.

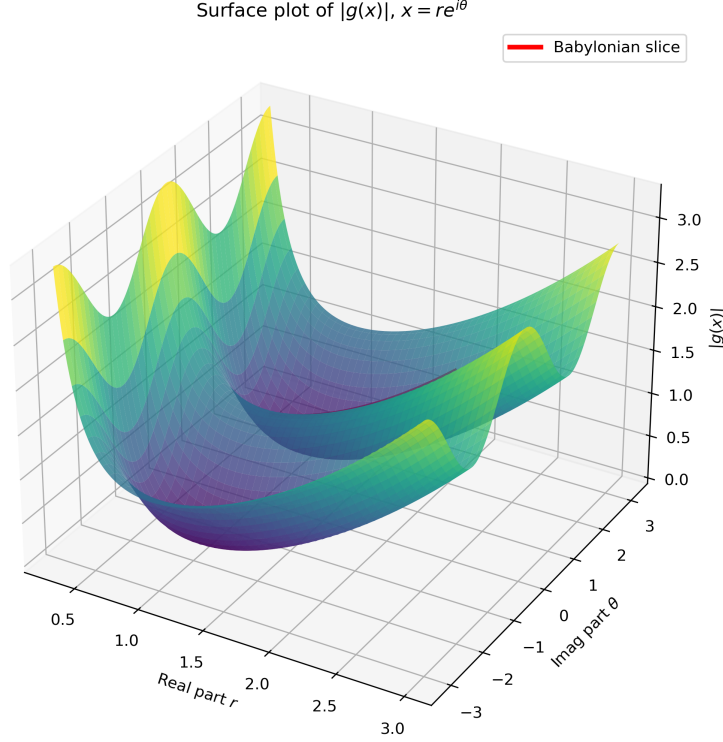


Figure 2: 3D surface plot of $|g(x)|$ generated via Python/Matplotlib. The valley along $|x| = 1$ corresponds to $\cos^2 \theta$. The Babylonian real slice $r > 1, \theta = 0$ runs along the front edge as a red curve. The surface exhibits conical/spiral geometry when viewed from different angles.

5 Discrete Sampling and Quantum Analogy

The selection of regular-number pairs (p, q) imposes a discrete set of admissible r -values. This discreteness mirrors the quantization of wave numbers in quantum mechanics. For example, the probability density of a particle in a one-dimensional infinite square well in a stationary state is proportional to $\sin^2(n\pi x/L)$ or $\cos^2(n\pi x/L)$ —precisely the form $\cos^2 \theta$ we obtained.

Moreover, the structure $e^{i\theta} + e^{-i\theta}$ represents the superposition of two counter-propagating waves. The square $|e^{i\theta} + e^{-i\theta}|^2$ gives an interference pattern. Plimpton 322's first column can therefore be seen as a discrete sampling of the intensity of such an interference pattern along a real parameter cut.

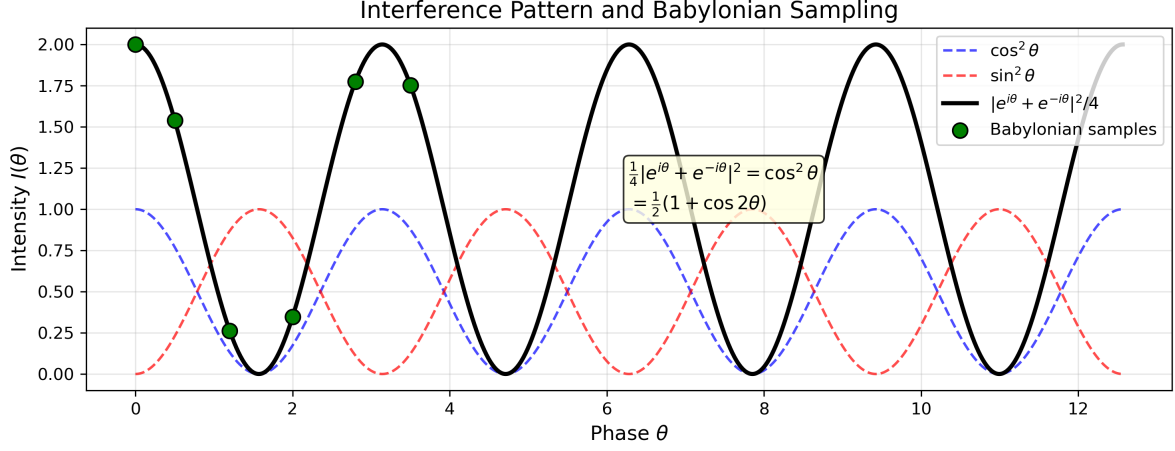


Figure 3: Interference pattern of two counter-propagating waves $e^{i\theta} + e^{-i\theta}$. The squared magnitude (black) equals $\cos^2 \theta$ (up to normalization), which decomposes as $\frac{1}{2}(1 + \cos 2\theta)$. Green dots indicate how Plimpton 322 samples this function at discrete θ values corresponding to regular p/q ratios. The complementary $\sin^2 \theta$ and $\cos^2 \theta$ functions sum to 1, reflecting probability conservation in quantum mechanics.

Mathematical Correspondence

The precise correspondence becomes clear when we note that the Babylonian generating function on the unit circle,

$$g(e^{i\theta}) = \cos^2 \theta,$$

is identical to the probability density of a quantum particle in certain superposition states. For instance, a particle in a symmetric double-well potential in the ground state superposition $\psi = \frac{1}{\sqrt{2}}(\psi_L + \psi_R)$ has probability density proportional to $\cos^2(kx)$. The discrete sampling of θ values in Plimpton 322 via p/q pairs thus resembles the discrete energy levels E_n in quantum systems, where only specific $\theta_n = \arccos(\sqrt{E_n/V_0})$ are allowed.

6 Historical and Philosophical Reflections

The Babylonians had no concept of complex numbers, analytic continuation, or quantum waves. Yet their empirically derived table encodes a function that, when viewed through modern mathematical lenses, reveals these deeper structures. This suggests that certain mathematical objects—like $g(x) = \frac{1}{4}(x + 1/x)^2$ —possess an inherent richness that transcends the historical context of their discovery. Plimpton 322 thus becomes more than a practical computational aid; it is a testament to the latent unity of mathematical truth across millennia.

7 Conclusion

We have shown that the generating function behind Plimpton 322, $g(r) = \frac{1}{4}(r + 1/r)^2$, harbors a hidden circular symmetry and, when extended to the complex plane, reduces

to $\cos^2 \theta$. This connection allows us to visualize the Babylonian data as points on a spiral surface and to draw analogies with quantum wave functions and interference patterns. While the ancient scribes could not have anticipated these interpretations, their work inadvertently prefigured mathematical structures that would later become central to theoretical physics. Re-examining ancient mathematical artifacts with modern tools can thus reveal surprising bridges between the distant past and contemporary science.

A Complete Data from Plimpton 322

Table 2: All 15 rows of Plimpton 322 with reconstructed p, q values

Row	$(c/b)^2$	Sexagesimal	p	q	a	b	c	
1	1.9834028	1;59,00,15	12	5	119	120	169	
2	1.9491586	1;56,56,58	64	27	3367	3456	4825	
3	1.9188021	1;55,07,41	75	32	4601	4800	6649	
4	1.8862479	1;53,10,29	125	54	12709	13500	18541	
5	1.8150077	1;48,54,01	9	4	65	72	97	
6	1.7851929	1;47,06,41	20	9	319	360	481	
7	1.7199837	1;43,11,56	54	25	2291	2700	3541	<i>Note: Sexagesimal</i>
8	1.6845877	1;41,33,45	32	15	799	960	1249	
9	1.6426694	1;38,33,36	25	12	481	600	769	
10	1.5861226	1;35,10,02	81	40	4961	6480	8161	
11	1.5625000	1;33,45	2	1	3	4	5	
12	1.4894168	1;29,21,54	48	25	1679	2400	2929	
13	1.4500174	1;27,00,03	15	8	161	240	289	
14	1.4302388	1;25,48,51	50	27	1771	2700	3229	
15	1.3871605	1;23,13,46	9	5	56	90	106	

notation uses semicolon for integer part and commas for fractional parts.

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